Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Joe Simoes & Sons Dairy Inc.

Physical address of dairy:

Number and Street	City	County	Zip Code
6900 Lambert RD	Elk Grove	Sacramento	95757

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1959

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X146-X004-X012-XXXX

B. OPERATORS

Simoes, Joe, Joey & Lester

Operator name: Simoes, Joe, Joey & Lester

Telephone no.: (123) 000-0000 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12055 Bruceville RD	Elk Grove	CA	95757

This operator is responsible for paying permit fees.

C. OWNERS

& Sons Dairy Inc, Joe Simoes

Legal owner name: & Sons Dairy Inc, Joe Simoes

Telephone no.: (123) 000-0000 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
12055 Bruceville RD	Elk Grove	CA	95757

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	75	0
Number under roof	900	125	0	0	0	0
Maximum number	900	125	0	0	75	0
Average number	900	125	0	0	75	0
Avg live weight (lbs)	1,400	1,500	0	0		

Predominant milk cow breed: Holstein

Average milk production: 70 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 25,197.82 *tons per reporting period*

Total nitrogen from manure: 328,487.09 *lbs per reporting period*

After ammonia losses (30% loss applied): 229,940.96 *lbs per reporting period*

Total phosphorus from manure: 54,382.31 *lbs per reporting period*

Total potassium from manure: 172,358.30 *lbs per reporting period*

Total salt from manure: 452,508.75 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 21,232,800 *gallons*

Total nitrogen generated: 50,879.51 *lbs*

Total phosphorus generated: 11,841.70 *lbs*

Total potassium generated: 76,851.03 *lbs*

Total salt generated: 506,769.49 *lbs*

	21,232,800	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	21,232,800	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #3	Ground water
Ag Well #4	Ground water
Ag Well #5	Ground water
Ag Well #7	Ground water
Ag Well #8	Ground water

Source Description	Type
Domestic Well #1	Ground water
Domestic Well #2	Ground water
Domestic Well #6	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	42	42	1	process wastewater	X146-X050-X001-XXXX
10	40	40	1	process wastewater	X146-X050-X009-XXXX
11	38	38	0	none	X146-X050-X065-XXXX
13	14	14	0	none	X146-X050-X019-XXXX
14	52	52	0	none	X146-X050-X019-XXXX
2	17	17	1	process wastewater	X146-X040-X012-XXXX
3	47	47	2	process wastewater	X146-X040-X012-XXXX
4	29	29	1	process wastewater	X146-X050-X012-XXXX
5	37	37	1	process wastewater	X146-X050-X012-XXXX
6	35	35	0	none	X146-X050-X065-XXXX
8	38	38	1	process wastewater	X146-X050-X065-XXXX
9	37	37	0	none	X146-X050-X065-XXXX
Totals for areas that were used for application	250	250	8		
Totals for areas that were not used for application	176	176	0		
Land application area totals	426	426	8		

B. CROPS AND HARVESTS

1

Field name: 1

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 42 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	1,260.00 *ton*	Dry-weight		70.3	8,200.00	1,890.00	9,160.00		5.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	146.12	33.68	163.23	1,030.00

10

10

Field name: 10

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 40 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	640.00 *ton*	Dry-weight		57.5	15,900.00	1,440.00	9,580.00		9.08

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	216.24	19.58	130.29	1,234.88

2

Field name: 2

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	510.00 *ton*	Dry-weight		64.6	15,400.00	2,290.00	9,370.00		5.94

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	327.10	48.64	199.02	1,261.66

3

Field name: 3

3

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 47 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	752.00 *ton*	Dry-weight		65.0	14,600.00	1,610.00	11,800.00		9.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	163.52	18.03	132.16	1,108.80

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 47 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	1,410.00 *ton*	Dry-weight		71.6	47,500.00	1,510.00	7,140.00		7.63

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	809.40	25.73	121.67	1,300.15

4

Field name: 4

06/04/2023: Corn, silage

Crop: Corn, silage Acres planted: 29 Plant date: 06/04/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	870.00 *ton*	Dry-weight		73.0	13,700.00	2,000.00	8,470.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	30.00	221.94	32.40	137.21	955.80

5

Field name: 5

10/15/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 37 Plant date: 10/15/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	592.00 *ton*	Dry-weight		62.6	17,400.00	1,700.00	11,600.00		11.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	208.24	20.35	138.83	1,424.19

8

Field name: 8

10/16/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 38 Plant date: 10/16/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/11/2023	608.00 *ton*	Dry-weight		59.4	15,700.00	1,460.00	10,500.00		9.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.00	203.97	18.97	136.42	1,273.22

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 06/03/2023: Corn, silage

Field name: 1

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	281.54	3,456,000.00 *gal*
Application event totals		0.00	0.00	0.00	281.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.95	7.36	57.08	334.75	842,400.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	197.08	2,419,200.00 *gal*
Application event totals		26.95	7.36	57.08	531.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.95	7.36	57.08	334.75	842,400.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	197.08	2,419,200.00 *gal*
Application event totals		26.95	7.36	57.08	531.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.95	7.36	57.08	334.75	842,400.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	197.08	2,419,200.00 *gal*
Application event totals		26.95	7.36	57.08	531.83	

1 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.95	7.36	57.08	334.75	842,400.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	197.08	2,419,200.00 *gal*
Application event totals		26.95	7.36	57.08	531.83	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.95	7.36	57.08	334.75	842,400.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	197.08	2,419,200.00 *gal*
Application event totals		26.95	7.36	57.08	531.83	

09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.61	9.01	63.05	362.65	702,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	197.08	2,419,200.00 *gal*
Application event totals		51.61	9.01	63.05	559.73	

10 - 10/16/2022: Oats, silage-soft dough

Field name: 10

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	94.02	20.91	111.42	861.15	842,400.00 *gal*
Ag Well #8	Ground water	0.00	0.00	0.00	181.69	2,419,200.00 *gal*
Application event totals		94.02	20.91	111.42	1,042.85	

2 - 06/03/2023: Corn, silage

2 - 06/03/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/03/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #4	Ground water	0.00	0.00	0.00	237.51	1,728,000.00 *gal*
Application event totals		0.00	0.00	0.00	237.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.90	9.27	71.81	421.18	429,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	158.34	1,152,000.00 *gal*
Application event totals		33.90	9.27	71.81	579.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.90	9.27	71.81	421.18	429,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	158.34	1,152,000.00 *gal*
Application event totals		33.90	9.27	71.81	579.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.90	9.27	71.81	421.18	429,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	158.34	1,152,000.00 *gal*
Application event totals		33.90	9.27	71.81	579.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.90	9.27	71.81	421.18	429,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	158.34	1,152,000.00 *gal*
Application event totals		33.90	9.27	71.81	579.52	

2 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	33.90	9.27	71.81	421.18	429,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	158.34	1,152,000.00 *gal*
Application event totals		33.90	9.27	71.81	579.52	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.92	13.60	95.19	547.53	429,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	158.34	1,152,000.00 *gal*
Application event totals		77.92	13.60	95.19	705.87	

3 - 10/15/2022: Oats, silage-soft dough

Field name: 3

Crop: Oats, silage-soft dough Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	80.02	17.80	94.83	732.90	842,400.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	176.11	2,419,200.00 *gal*
Application event totals		80.02	17.80	94.83	909.01	

3 - 06/04/2023: Corn, silage

Field name: 3

Crop: Corn, silage Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

3 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #3	Ground water	0.00	0.00	0.00	272.55	3,744,000.00 *gal*
Application event totals		0.00	0.00	0.00	272.55	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.30	6.09	47.23	276.98	780,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	174.71	2,400,000.00 *gal*
Application event totals		22.30	6.09	47.23	451.70	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.30	6.09	47.23	276.98	780,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	174.71	2,400,000.00 *gal*
Application event totals		22.30	6.09	47.23	451.70	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.30	6.09	47.23	276.98	780,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	174.71	2,400,000.00 *gal*
Application event totals		22.30	6.09	47.23	451.70	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.30	6.09	47.23	276.98	780,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	174.71	2,400,000.00 *gal*
Application event totals		22.30	6.09	47.23	451.70	

3 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.30	6.09	47.23	276.98	780,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	174.71	2,400,000.00 *gal*
Application event totals		22.30	6.09	47.23	451.70	

09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	51.24	8.95	62.60	360.08	780,000.00 *gal*
Ag Well #3	Ground water	0.00	0.00	0.00	174.71	2,400,000.00 *gal*
Application event totals		51.24	8.95	62.60	534.79	

4 - 06/04/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/04/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #4	Ground water	0.00	0.00	0.00	185.64	2,304,000.00 *gal*
Application event totals		0.00	0.00	0.00	185.64	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.30	6.91	53.58	314.23	546,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	123.76	1,536,000.00 *gal*
Application event totals		25.30	6.91	53.58	437.99	

4 - 06/04/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.30	6.91	53.58	314.23	546,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	123.76	1,536,000.00 *gal*
Application event totals		25.30	6.91	53.58	437.99	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.30	6.91	53.58	314.23	546,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	123.76	1,536,000.00 *gal*
Application event totals		25.30	6.91	53.58	437.99	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.30	6.91	53.58	314.23	546,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	123.76	1,536,000.00 *gal*
Application event totals		25.30	6.91	53.58	437.99	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.30	6.91	53.58	314.23	546,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	123.76	1,536,000.00 *gal*
Application event totals		25.30	6.91	53.58	437.99	

09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	58.13	10.15	71.02	408.50	546,000.00 *gal*
Ag Well #4	Ground water	0.00	0.00	0.00	123.76	1,536,000.00 *gal*
Application event totals		58.13	10.15	71.02	532.26	

5 - 10/15/2022: Oats, silage-soft dough

5 - 10/15/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 10/15/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	101.65	22.61	120.46	930.98	842,400.00 *gal*
Ag Well #5	Ground water	0.00	0.00	0.00	130.95	2,419,200.00 *gal*
Application event totals		101.65	22.61	120.46	1,061.93	

8 - 10/16/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 10/16/2022

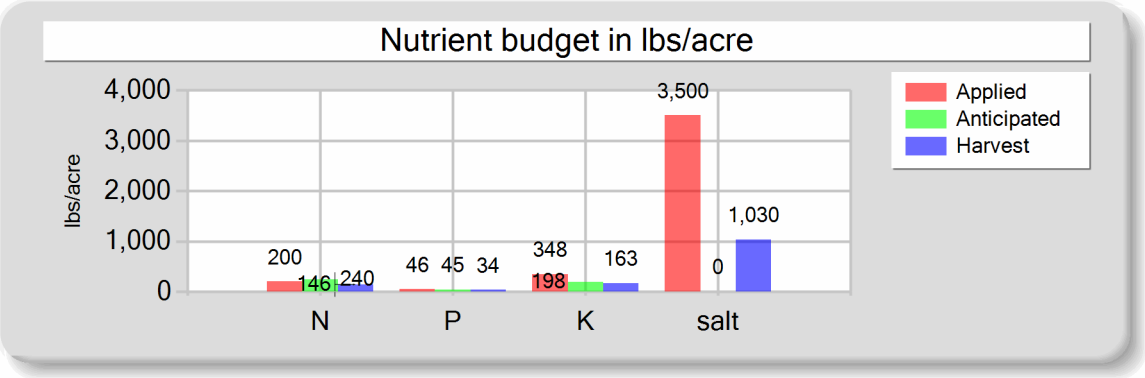
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	98.97	22.01	117.29	906.48	842,400.00 *gal*
Lagoon Water	Process wastewater	284.23	63.22	336.82	2,603.22	2,419,200.00 *gal*
Application event totals		383.20	85.24	454.11	3,509.70	

B. NUTRIENT BUDGET

1 - 06/03/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	186.35	45.83	348.42	2,036.42
Fresh water	0.00	0.00	0.00	1,463.99
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	200.35	45.83	348.42	3,500.41
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	146.12	33.68	163.23	1,030.00
Nutrient balance	54.22	12.15	185.19	2,470.41
Applied to removed ratio	1.37	1.36	2.13	3.40

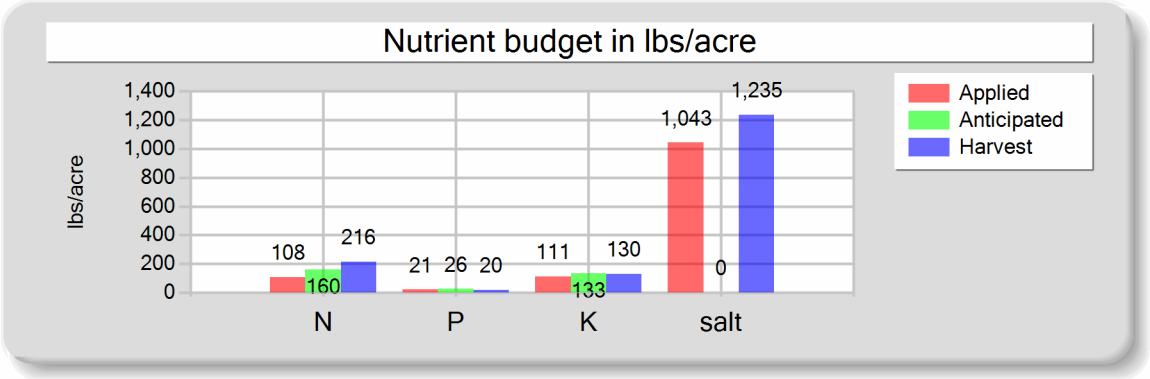
Fresh water applied	
17,971,200.00	*gallons*
661.82	*acre-inches*
15.76	*inches/acre*

Process wastewater applied	
4,914,000.00	*gallons*
180.97	*acre-inches*
4.31	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 10/16/2022: Oats, silage-soft dough

Field name: 10 Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	94.02	20.91	111.42	861.15
Fresh water	0.00	0.00	0.00	181.69
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	108.02	20.91	111.42	1,042.85
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	216.24	19.58	130.29	1,234.88
Nutrient balance	-108.22	1.33	-18.87	-192.03
Applied to removed ratio	0.50	1.07	0.86	0.84

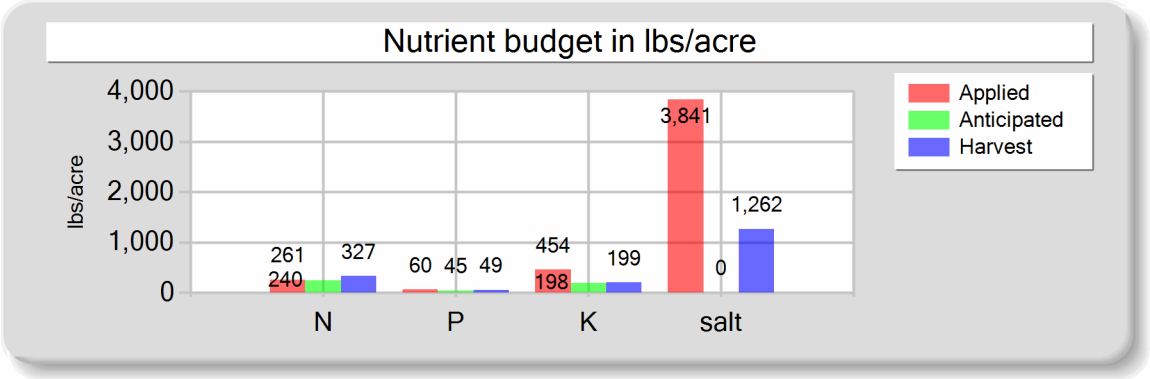
Fresh water applied	
2,419,200.00	*gallons*
89.09	*acre-inches*
2.23	*inches/acre*

Process wastewater applied	
842,400.00	*gallons*
31.02	*acre-inches*
0.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 06/03/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	247.44	59.93	454.24	2,653.42
Fresh water	0.00	0.00	0.00	1,187.54
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	261.44	59.93	454.24	3,840.96
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	327.10	48.64	199.02	1,261.66
Nutrient balance	-65.65	11.29	255.22	2,579.30
Applied to removed ratio	0.80	1.23	2.28	3.04

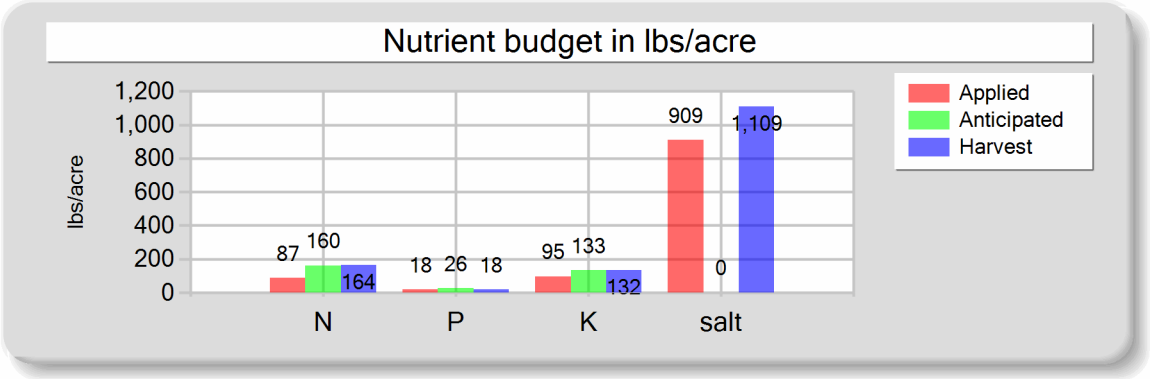
Fresh water applied	
8,640,000.00	*gallons*
318.18	*acre-inches*
18.72	*inches/acre*

Process wastewater applied	
2,574,000.00	*gallons*
94.79	*acre-inches*
5.58	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 10/15/2022: Oats, silage-soft dough

Field name: 3 Crop: Oats, silage-soft dough Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	80.02	17.80	94.83	732.90
Fresh water	0.00	0.00	0.00	176.11
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	87.02	17.80	94.83	909.01
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	163.52	18.03	132.16	1,108.80
Nutrient balance	-76.50	-0.23	-37.33	-199.79
Applied to removed ratio	0.53	0.99	0.72	0.82

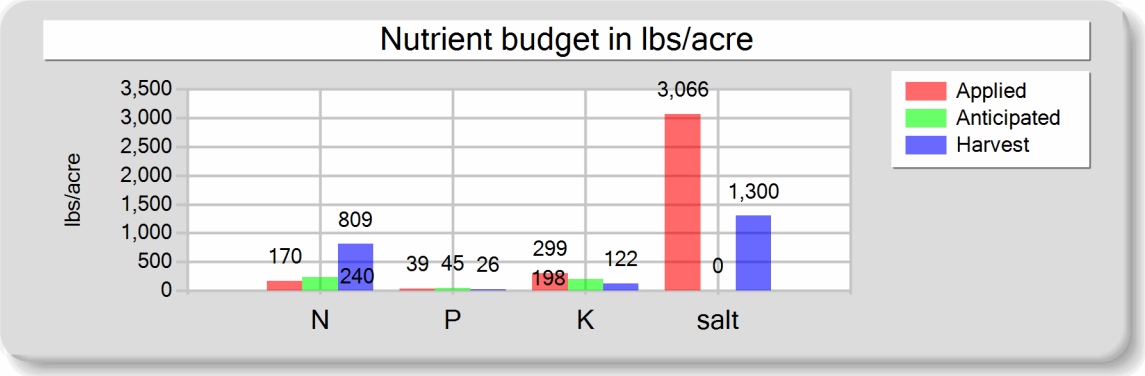
Fresh water applied	
2,419,200.00	*gallons*
89.09	*acre-inches*
1.90	*inches/acre*

Process wastewater applied	
842,400.00	*gallons*
31.02	*acre-inches*
0.66	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 06/04/2023: Corn, silage

Field name: 3 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	162.73	39.41	298.73	1,744.99
Fresh water	0.00	0.00	0.00	1,320.83
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	169.73	39.41	298.73	3,065.82
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	809.40	25.73	121.67	1,300.15
Nutrient balance	-639.67	13.68	177.06	1,765.67
Applied to removed ratio	0.21	1.53	2.46	2.36

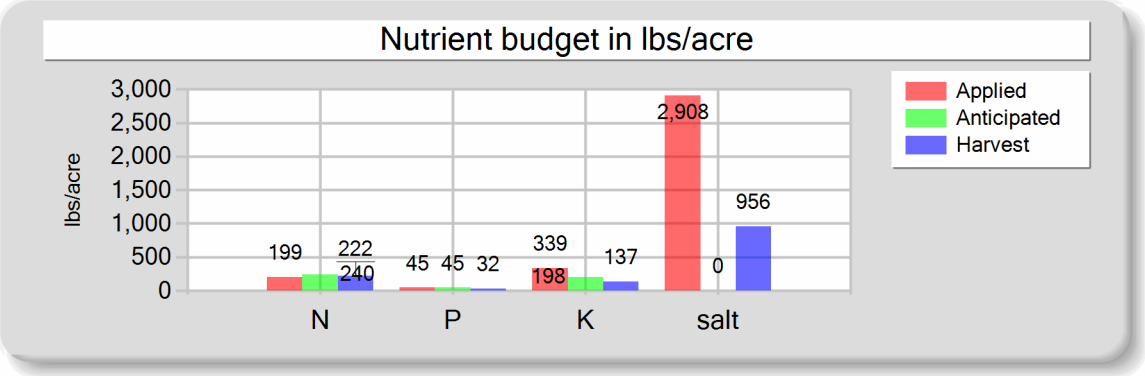
Fresh water applied	
18,144,000.00	*gallons*
668.18	*acre-inches*
14.22	*inches/acre*

Process wastewater applied	
4,680,000.00	*gallons*
172.35	*acre-inches*
3.67	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/04/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/04/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	184.61	44.72	338.90	1,979.66
Fresh water	0.00	0.00	0.00	928.19
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	198.61	44.72	338.90	2,907.86
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	221.94	32.40	137.21	955.80
Nutrient balance	-23.33	12.32	201.69	1,952.06
Applied to removed ratio	0.89	1.38	2.47	3.04

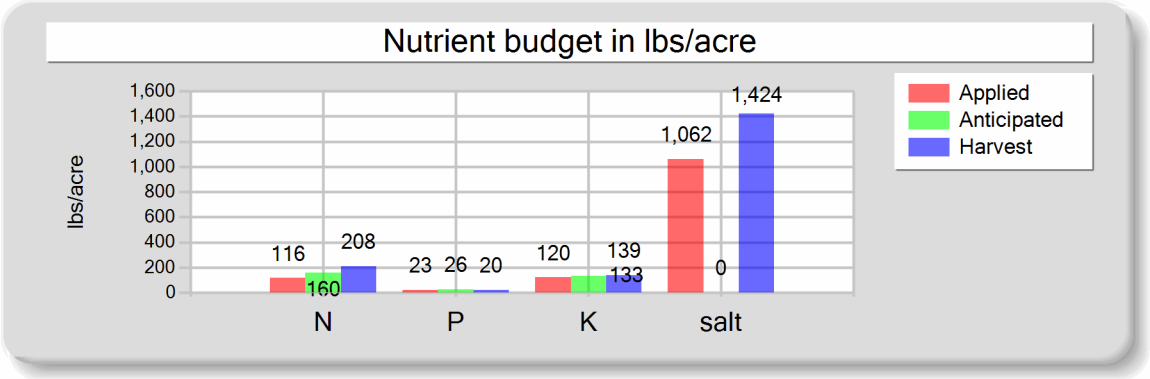
Fresh water applied	
11,520,000.00	*gallons*
424.24	*acre-inches*
14.63	*inches/acre*

Process wastewater applied	
3,276,000.00	*gallons*
120.64	*acre-inches*
4.16	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 10/15/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 10/15/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	101.65	22.61	120.46	930.98
Fresh water	0.00	0.00	0.00	130.95
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	115.65	22.61	120.46	1,061.93
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	208.24	20.35	138.83	1,424.19
Nutrient balance	-92.60	2.26	-18.37	-362.26
Applied to removed ratio	0.56	1.11	0.87	0.75

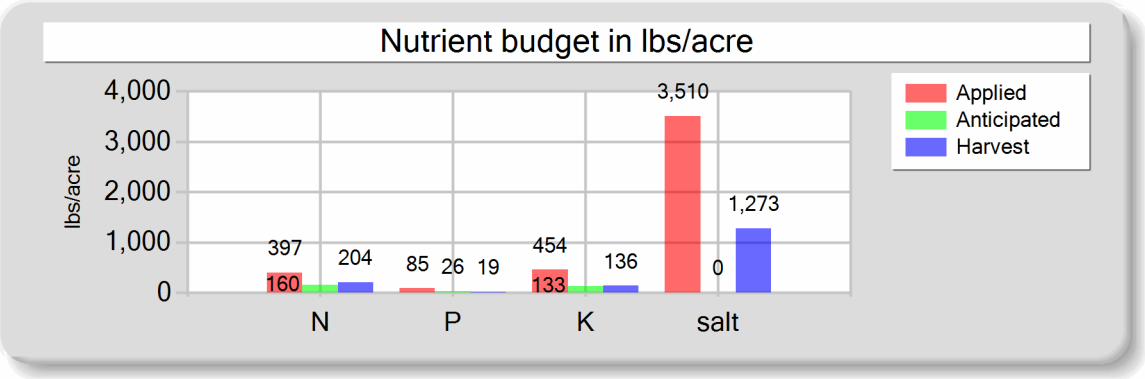
Fresh water applied	
2,419,200.00	*gallons*
89.09	*acre-inches*
2.41	*inches/acre*

Process wastewater applied	
842,400.00	*gallons*
31.02	*acre-inches*
0.84	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 10/16/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 10/16/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	383.20	85.24	454.11	3,509.70
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	397.20	85.24	454.11	3,509.70
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	203.97	18.97	136.42	1,273.22
Nutrient balance	193.23	66.27	317.70	2,236.48
Applied to removed ratio	1.95	4.49	3.33	2.76

Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
3,261,600.00 *gallons*
120.11 *acre-inches*
3.16 *inches/acre*

Total harvests for the crop
1 *harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Manure 2310009-02

Sample and source description: Manure 2310009-02

Sample date: 09/28/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,700.00	2,540.00	4,970.00							43.60
DL	0.50	0.50	0.50							0.10

B. PROCESS WASTEWATER ANALYSES

Lagoon Water 2301427-01

Sample and source description: Lagoon Water 2301427-01

Sample date: 01/27/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	535.00	228.00			119.00	634.00								4,600.00	4,900
DL	0.60	6.00			0.10	10.00								1.00	1

Lagoon Water 2303486-01

Sample and source description: Lagoon Water 2303486-01

Sample date: 03/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	456.00	21.80			7.60	47.20								4,310.00	3,000
DL	0.60	0.60			0.10	1.00								1.00	1

Lagoon Water 2307293-01

Sample and source description: Lagoon Water 2307293-01

Sample date: 07/19/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	161.00	157.00			44.00	341.00								3,110.00	2,000
DL	0.60	0.60			0.10	10.00								1.00	1

Lagoon Water 2310009-01

Sample and source description: Lagoon Water 2310009-01

Sample date: 09/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	370.00	256.00			64.60	452.00								4,340.00	2,600
DL	0.60	0.60			0.10	10.00								1.00	1

C. FRESH WATER ANALYSES

Ag Well #3

2312526-04

Sample description: 2312526-04

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											374.00	410
DL											1.00	1

Ag Well #4

Ag Well #4

2312526-05

Sample description: 2312526-05

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											441.00	280
DL											1.00	1

Ag Well #5

2312526-06

Sample description: 2312526-06

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											329.00	240
DL											1.00	1

Ag Well #7

2312526-07

Sample description: 2312526-07

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											305.00	170
DL											1.00	1

Ag Well #8

Ag Well #8

2312526-08

Sample description: 2312526-08

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											536.00	360
DL											1.00	1

Domestic Well #1

2312526-01

Sample description: 2312526-01

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.20										1,260.00	
DL	0.40										1.00	

Domestic Well #2

2312526-02

Sample description: 2312526-02

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	10.00										1,300.00	
DL	0.40										1.00	

Domestic Well #6

Domestic Well #6

2312526-03

Sample description: 2312526-03

Sample date: 12/28/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value											1,350.00	
DL											1.00	

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 06/03/2023: Corn, silage

2310008-01

Sample and source description: 2310008-01

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,200.00	1,890.00	9,160.00		5.78
DL	0.50	0.50	0.50		0.10

10 - 10/16/2022: Oats, silage-soft dough

10 - 10/16/2022: Oats, silage-soft dough

2305224-04

Sample and source description: 2305224-04

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	1,440.00	9,580.00		9.08
DL	1.00	0.50	0.50		0.10

2 - 06/03/2023: Corn, silage

2310008-02

Sample and source description: 2310008-02

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,400.00	2,290.00	9,370.00		5.94
DL	0.50	0.50	0.50		0.10

3 - 10/15/2022: Oats, silage-soft dough

2305224-01

Sample and source description: 2305224-01

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,600.00	1,610.00	11,800.00		9.90
DL	1.00	0.50	0.50		0.10

3 - 06/04/2023: Corn, silage

2310008-03

Sample and source description: 2310008-03

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	47,500.00	1,510.00	7,140.00		7.63
DL	1.00	0.50	0.50		0.10

4 - 06/04/2023: Corn, silage

2310008-04

Sample and source description: 2310008-04

Sample date: 09/29/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,700.00	2,000.00	8,470.00		5.90
DL	0.50	0.50	0.50		0.10

5 - 10/15/2022: Oats, silage-soft dough

2305224-02

Sample and source description: 2305224-02

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 62.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,400.00	1,700.00	11,600.00		11.90
DL	1.00	0.50	0.50		0.10

8 - 10/16/2022: Oats, silage-soft dough

2305224-03

Sample and source description: 2305224-03

Sample date: 05/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,700.00	1,460.00	10,500.00		9.80
DL	0.50	0.50	0.50		0.10

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

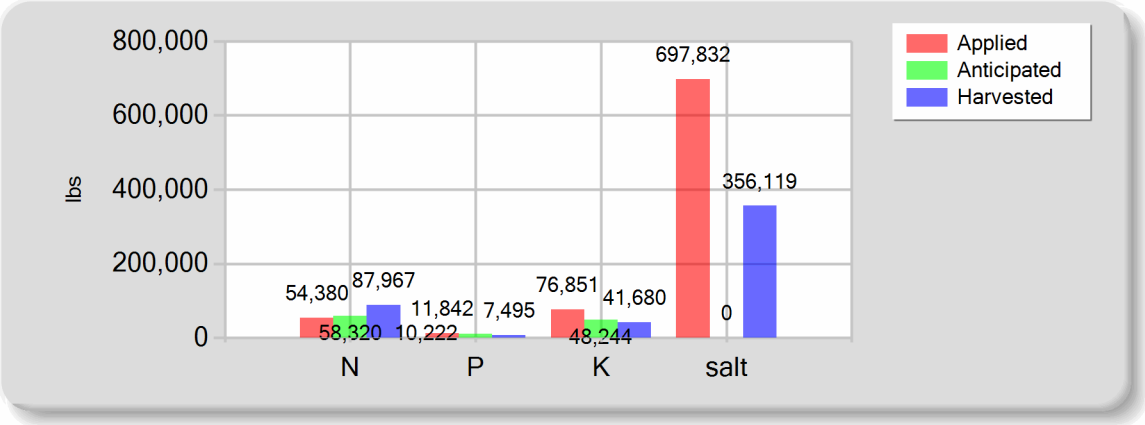
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

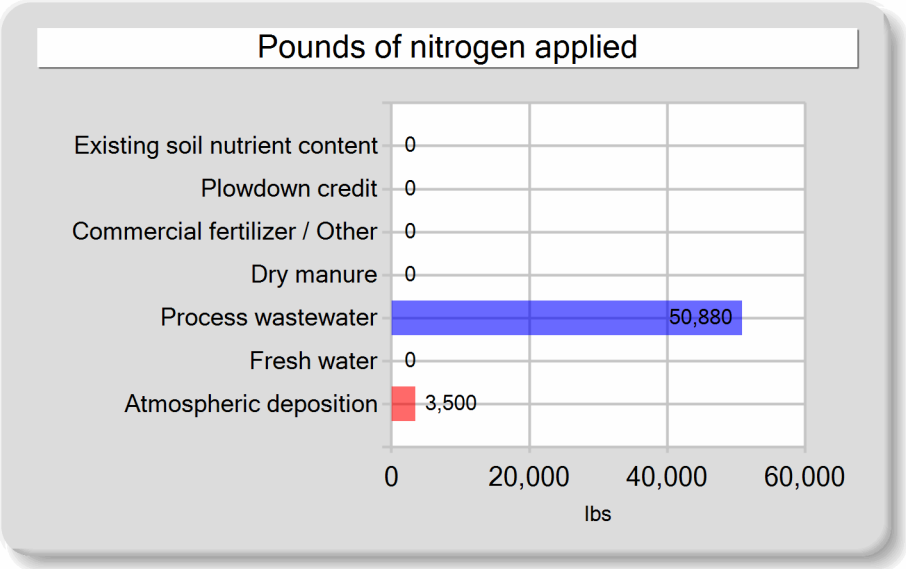
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

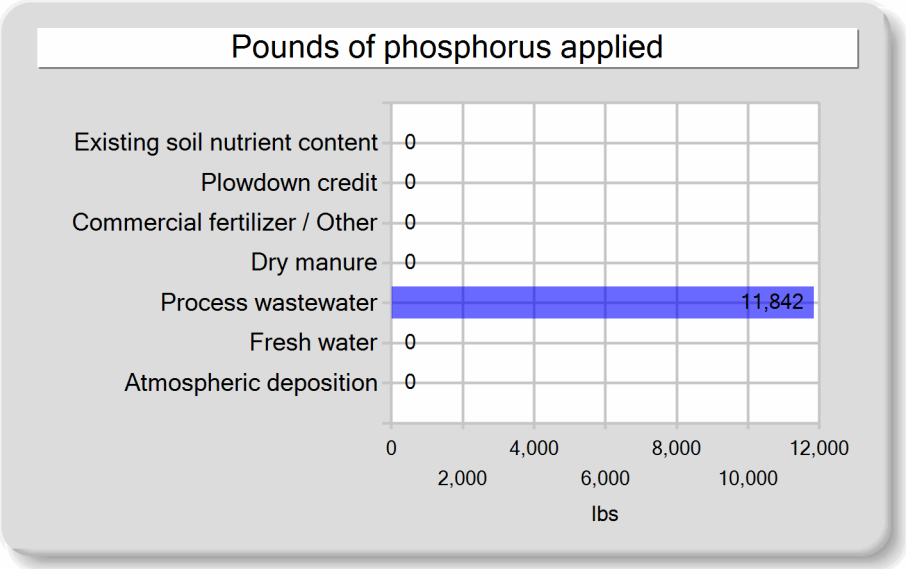
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	50,879.51	11,841.70	76,851.03	506,769.49
Fresh water	0.00	0.00	0.00	191,062.31
Atmospheric deposition	3,500.00	0.00	0.00	0.00
Total nutrients applied	54,379.51	11,841.70	76,851.03	697,831.80
Anticipated crop nutrient removal	58,320.00	10,222.20	48,243.60	0.00
Actual crop nutrient removal	87,966.97	7,494.80	41,680.03	356,119.44
Nutrient balance	-33,587.46	4,346.90	35,170.99	341,712.36
Applied to removed ratio	0.62	1.58	1.84	1.96

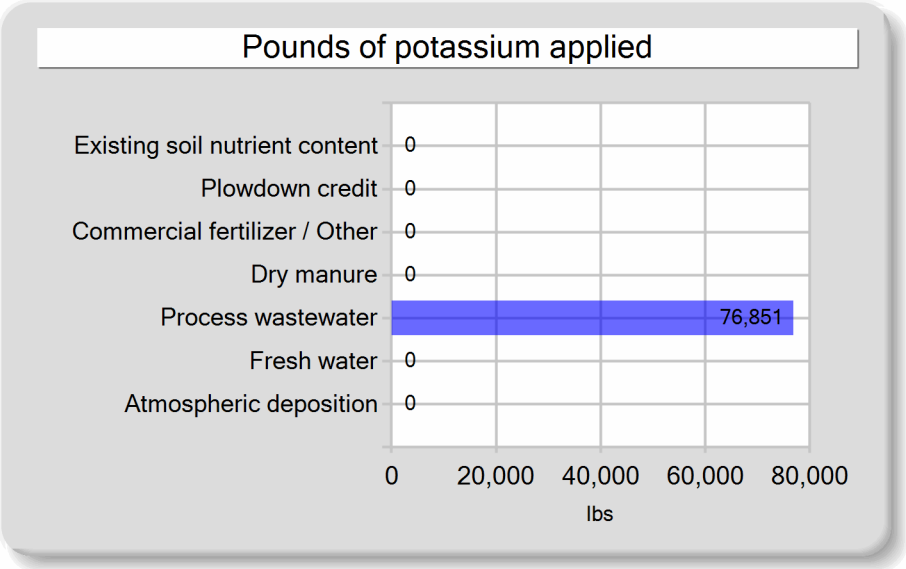
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

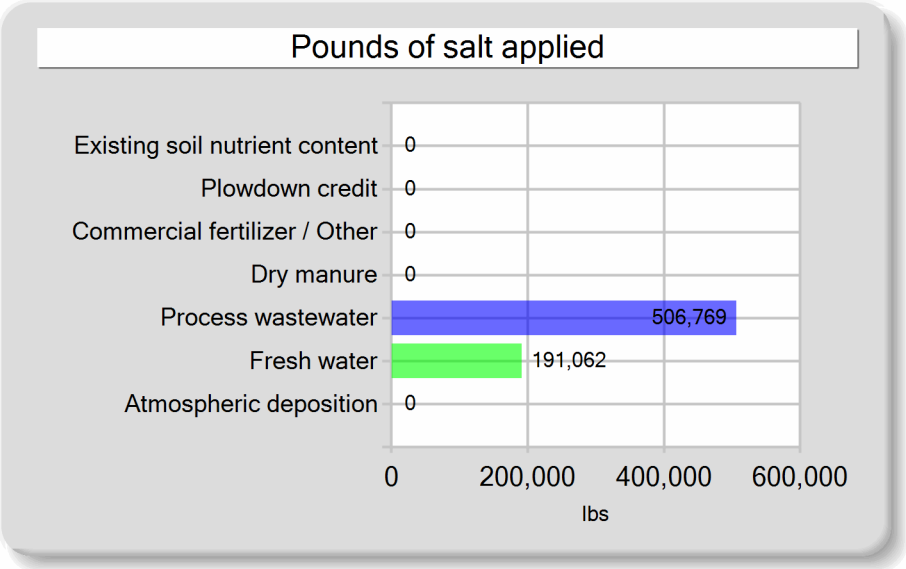


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	No
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Joe Simoes & Sons Dairy Inc	Joe, Joey & Lester Simoes
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Joey Simoes 7-17-2024	7-17-2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment	N/A
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .	

Manure/Process Wastewater Tracking Manifests	N/A
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.	

Corrective Actions Documents	N/A
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.	

Groundwater Monitoring	N/A
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.	

Storm Water Monitoring	N/A
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.	